

SAULT SAINTE MARIE, MI, USA

WMO: 727340

Lat: 46.479N

Lon: 84.357W

Elev: 722

StdP: 14.32

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14847

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-4.9	-18.8	2.0	-9.4	-13.2	2.8	-3.5	24.0	21.9	20.8	21.1	5.2	80	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	83.7	69.5	80.7	67.8	77.9	66.1	72.2	80.2	70.2	77.3	68.4	75.1	9.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.4	111.2	76.4	67.6	104.2	74.2	65.8	97.7	72.1	36.2	80.0	34.6	77.1	33.0	75.4	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.7	18.5	16.6	DB	-16.9	88.6	6.6	2.9	-21.7	90.7	-25.6	92.4	-29.3	94.0	-34.2	96.1	
			WB	-17.3	75.9	6.6	2.2	-22.1	77.5	-25.9	78.7	-29.6	80.0	-34.4	81.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.2	15.6	17.6	26.5	38.9	51.6	60.9	65.8	65.2	58.6	46.4	34.5	23.5
	DBStd	19.73	11.82	10.77	10.66	8.20	7.88	6.63	5.82	5.28	7.33	7.64	8.68	10.16
	HDD50	4593	1067	907	731	343	75	4	0	0	12	167	467	820
	HDD65	8532	1532	1327	1194	784	421	155	61	63	217	579	914	1285
	CDD50	1755	0	0	2	9	124	331	491	472	269	54	3	0
	CDD65	219	0	0	0	0	6	32	87	69	24	1	0	0
	CDH74	1662	0	0	1	3	81	323	686	429	135	4	0	0
	CDH80	325	0	0	0	0	13	62	155	74	21	0	0	0
Wind	WSAvg	7.4	7.5	7.9	7.9	8.4	8.0	6.7	6.6	6.4	6.6	7.6	8.1	7.8
Precipitation	PrecAvg	34.0	2.3	1.6	2.0	2.4	2.7	2.9	2.9	3.4	3.7	3.7	3.5	2.8
	PrecMax	48.8	4.5	3.7	5.0	5.5	7.4	7.4	7.2	9.5	8.3	7.6	7.7	6.3
	PrecMin	22.4	0.5	0.2	0.2	0.9	0.6	0.4	0.6	1.0	0.5	0.2	0.9	0.6
	PrecStd	5.6	0.9	0.8	1.1	1.0	1.3	1.4	1.4	1.5	1.8	1.7	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	43.5	58.5	68.5	81.6	85.6	88.2	85.8	83.2	72.5	59.8	46.4
		MCWB	38.7	38.9	48.2	52.9	65.7	71.1	72.0	71.2	70.4	62.9	53.0	43.9
	2%	DB	37.2	38.4	49.7	62.0	75.5	81.4	84.2	81.9	77.6	66.6	53.0	41.5
		MCWB	35.4	35.2	42.2	49.0	60.9	67.7	69.7	69.2	66.9	59.3	48.6	39.3
	5%	DB	34.8	35.5	44.4	56.4	70.9	77.6	81.0	78.8	74.1	62.1	49.3	38.6
		MCWB	33.1	33.0	39.0	45.3	57.8	64.7	67.8	67.3	65.1	56.7	45.6	36.6
	10%	DB	32.3	32.8	40.0	51.6	66.1	74.2	77.8	75.7	70.5	57.9	46.2	36.4
		MCWB	30.6	30.6	35.6	43.0	54.9	63.0	66.4	65.8	63.2	53.4	43.2	34.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.2	39.8	50.7	56.2	68.2	74.0	75.2	74.1	71.8	65.3	53.9	44.8
		MCDB	40.2	42.1	55.2	65.8	78.2	83.2	83.6	81.2	79.5	68.6	57.8	46.2
	2%	WB	35.6	35.5	43.9	50.7	63.7	69.8	72.4	71.4	69.1	60.7	49.6	39.6
		MCDB	36.8	37.8	48.8	58.2	72.4	78.0	80.8	78.3	75.0	65.8	52.9	41.1
	5%	WB	33.3	33.2	39.3	47.0	59.9	67.2	70.2	69.4	66.9	57.2	46.1	37.0
		MCDB	34.7	35.0	43.5	54.8	67.4	74.9	77.6	75.7	72.2	61.3	48.8	38.3
	10%	WB	30.6	30.6	36.0	43.8	56.4	64.7	68.2	67.6	64.5	54.0	43.6	34.7
		MCDB	32.3	32.7	39.6	50.9	64.2	71.6	75.1	73.8	69.3	57.7	46.3	36.1
Mean Daily Temperature Range	5% DB	MDBR	13.9	14.9	15.6	16.1	19.9	20.3	19.3	17.9	17.2	13.7	11.2	11.6
		MCDBR	11.4	14.2	18.6	25.7	27.7	25.6	23.9	21.8	20.8	19.3	15.3	11.5
		MCWBR	10.7	12.2	12.7	15.3	15.6	14.1	12.1	11.1	11.3	12.9	12.0	10.7
	5% WB	MCDBR	11.1	13.7	16.6	22.7	24.2	21.8	20.2	18.0	18.6	17.3	14.1	10.8
		MCWBR	10.8	12.5	12.3	15.1	15.2	13.6	11.9	10.7	11.6	12.6	12.0	10.5
Clear-Sky Solar Irradiance	taub	0.278	0.289	0.318	0.355	0.368	0.386	0.387	0.377	0.363	0.329	0.301	0.285	
	taud	2.336	2.293	2.328	2.331	2.376	2.347	2.347	2.362	2.388	2.426	2.507	2.508	2.367
	Ebn at Noon	260	280	288	286	285	279	277	275	270	265	253	243	
	Edh at Noon	23	30	34	37	37	38	37	35	31	24	20	21	
All-Sky Solar Radiation	RadAvg	346	632	1041	1415	1762	1898	1904	1610	1206	704	371	269	
	RadStd	38	70	116	158	145	131	107	106	93	80	47	24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+216	N/A
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page